

Filip Tomaska

Santa Barbara, CA
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Education

PhD in Dynamical Neuroscience 2021–present
University of California, Santa Barbara

MSc in Neurobiology 2016–2018
Charles University, Faculty of Science
Thesis: The role of inhibitory interneurons in encoding of complex sounds by the mouse auditory cortex
Grade 1.0 (scale 4–1 [low – high]), summa cum laude

BSc in Biology 2013–2016
Charles University, Faculty of Science
Thesis: Modern optical in vivo methods in neurophysiological research
Grade 1.47 (scale 4–1 [low – high])

Research Experience

Smith Lab, UCSB 2021–present
Graduate student. Surgical preparations and large-scale calcium imaging.

Schreiter Lab, HHMI 2020–2021
Research technician (full-time). Surgical preparation and in vivo screening of sensors; software and hardware development for novel imaging techniques.

Podgorski/Schreiter Lab, HHMI 2019–2020
Joint visiting project. Surgical preparation of mice and in vivo imaging in support of sensor validation.

Novak Lab, Charles University 2018–2020
Development of novel surgical preparations and imaging techniques.

Turecek Lab, Czech Academy of Sciences 2016–2019
Surgical preparation and in vivo imaging supporting auditory neuroscience research.

Teaching Experience, UCSB

- **ECE 6**, The Physics of Energy, Information, and Communication 2024
- **PSY 10A**, Research Methods in Psychological and Brain Sciences 2025, 2026
- **PSY 111**, Fundamentals of Behavioral Neuroscience 2026

Honours and Awards

- MSc summa cum laude 2018
- Best Bachelor's Thesis, Department of Physiology 2016
- Academic Scholarship 2014, 2016

Grants

Grant Agency of Charles University

2016–2019

Role of inhibitory interneurons in attractor categorization of complex stimuli in mouse auditory cortex.

The Sculpted Light in the Brain Association

2024

Travel Grant for Sculpted Light in the Brain 2024.

Personal Skills

Languages: Slovak (native), English (Proficient : TOEFL 116), French (C1)

Programming: MATLAB, Python

Publications

Yu, C.-H., Yu, Y., Canzano, J. S., Fei, Y., **Tomaska, F.**, & Smith, S. L. (2026). Non-inertial scan angle multiplier for expanded fields-of-view. *Optica*, 13(3), 409. doi.org/10.1364/OPTICA.570761

Farrants, H., Shuai, Y., Lemon, W. C., Monroy Hernandez, C., Zhang, D., Yang, S., Patel, R., Qiao, G., Frei, M. S., Plutkis, S. E., Grimm, J. B., Hanson, T. L., **Tomaska, F.**, Turner, G. C., Stringer, C., Keller, P. J., Beyene, A. G., Chen, Y., Liang, Y., Lavis, L. D., & Schreiter, E. R. (2024). A modular chemigenetic calcium indicator for multiplexed in vivo functional imaging. *Nature Methods*, 21(10), 1916–1925. doi.org/10.1038/s41592-024-02411-6

Aggarwal, A., Liu, R., Chen, Y., Ralowicz, A. J., Bergerson, S. J., **Tomaska, F.**, Mohar, B., Hanson, T. L., Hasseman, J. P., Reep, D., Tsegaye, G., Yao, P., Ji, X., Kloos, M., Walpita, D., Patel, R., Mohr, M. A., Tillberg, P. W., GENIE Project Team, Looger, L. L., Marvin, J. S., Hoppa, M. B., Konnerth, A., Kleinfeld, D., Schreiter, E. R., & Podgorski, K. (2023). Glutamate indicators with improved activation kinetics and localization for imaging synaptic transmission. *Nature Methods*, 20(6), 925–934. doi.org/10.1038/s41592-023-01863-6

Chvojka, J., Kudlacek, J., Chang, W.-C., Novak, O., **Tomaska, F.**, Otahal, J., Jefferys, J. G. R., & Jiruska, P. (2019). The role of interictal discharges in ictogenesis — a dynamical perspective. *Epilepsy & Behavior*. doi.org/10.1016/j.yebeh.2019.106591

References

Available upon request.